Grade 9 Mathematics – Study Notes

These notes follow the standard Grade 9 CAPS content. Check them against your school's exam guidelines.

Numbers, exponents and scientific notation

Number systems

Set	Symbol	Contains	Examples
Natural numbers	$\mathbb{N}$	Counting numbers	1; 2; 3; …
Whole numbers	$\mathbb{N}_0$	Natural numbers and 0	0; 1; 2; 3; …
Integers	$\mathbb{Z}$	Whole numbers and their negatives	…; −2; −1; 0; 1; 2; …
Rational numbers	$\mathbb{Q}$	Can be written as a fraction a/b (b ≠ 0); terminating or recurring decimals	½; −3; 0.75; 0.333…
Irrational numbers	$\mathbb{Q}'$	Cannot be written as a fraction; non-recurring, non-terminating decimals	√2; √5; π
Real numbers	$\mathbb{R}$	All rational and irrational numbers	

- **Prime number:** exactly two factors, 1 and itself (2, 3, 5, 7, 11…). 1 is not prime; 2 is the only even prime.
- **HCF** (highest common factor) and **LCM** (lowest common multiple): write each number as a product of prime factors. HCF = common primes (lowest powers); LCM = all primes (highest powers).
  - $24 = 2^3 \times 3$ and $36 = 2^2 \times 3^2$ → HCF = $2^2 \times 3$ = **12**; LCM = $2^3 \times 3^2$ = **72**.

Integers

- Adding a negative = subtracting: 5 + (−3) = 2. Subtracting a negative = adding: 5 − (−3) = 8.
- Multiplying/dividing: same signs → **positive**; different signs → **negative**. (−4) × (−3) = 12; (−12) ÷ 4 = −3.
- **Order of operations (BODMAS):** Brackets, Of/Orders (powers, roots), Division and Multiplication (left to right), Addition and Subtraction (left to right).
  - $3 + 4 \times 2^2 = 3 + 4 \times 4 = 3 + 16$ = **19**

Exponents

In a^n, *a* is the **base** and *n* is the **exponent**: $2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$.

Law	Rule	Example
Multiply same base	$a^m \times a^n = a^{m+n}$	$x^3 \times x^4 = x^7$
Divide same base	$a^m \div a^n = a^{m-n}$	$y^8 \div y^3 = y^5$
Power of a power	$(a^m)^n = a^{mn}$	$(x^2)^3 = x^6$

Law	Rule	Example
Power of a product	$(ab)^n = a^n b^n$	$(2x)^3 = 8x^3$
Power of a quotient	$(a/b)^n = a^n/b^n$	$(x/3)^2 = x^2/9$
Zero exponent	$a^0 = 1$ ($a \neq 0$)	$5^0 = 1$; $(3xy)^0 = 1$
Negative exponent	$a^{-n} = 1/a^n$	$2^{-3} = 1/8$

- Watch the signs: $(-2)^4 = 16$ but $-2^4 = -16$; $(-2)^3 = -8$.
- Simplify coefficients and variables separately: $3x^2y \times 4x^5y^3 = 12x^7y^4$; $(2a^3)^4 = 16a^{12}$; $12x^6 \div 4x^2 = 3x^4$.

Squares, cubes and roots

- $\sqrt{49} = 7$ because $7^2 = 49$; $\sqrt[3]{64} = 4$ because $4^3 = 64$; $\sqrt[3]{(-27)} = -3$.
- Learn squares to $15^2 = 225$ and cubes to $6^3 = 216$.
- $\sqrt{(a \times b)} = \sqrt{a} \times \sqrt{b}$, e.g. $\sqrt{(16 \times 9)} = 4 \times 3 = 12$. But $\sqrt{(9 + 16)} = \sqrt{25} = 5$, **not** $3 + 4$.

Scientific notation

A number written as $\mathbf{a \times 10^n}$, where $\mathbf{1 \leq a < 10}$ and n is an integer.

- Large numbers → positive exponent: 45 000 000 = $\mathbf{4.5 \times 10^7}$ (decimal moved 7 places left).
- Small numbers → negative exponent: 0.00032 = $\mathbf{3.2 \times 10^{-4}}$ (decimal moved 4 places right).
- Multiply: $(3 \times 10^4) \times (2 \times 10^3) = 6 \times 10^7$.
- Divide: $(8 \times 10^9) \div (2 \times 10^3) = 4 \times 10^6$.
- Adjust if a is not between 1 and 10: $25 \times 10^5 = 2.5 \times 10^6$.

Algebraic expressions, factorising and equations

Terms and vocabulary

In $\mathbf{3x^2 - 5x + 7}$:

- There are **3 terms** (separated by + and −), so it is a **trinomial**. (1 term = monomial, 2 terms = binomial.)
- **Coefficient** of x^2 is 3; coefficient of x is −5; **constant** is 7.
- **Degree** = highest exponent = 2.
- **Like terms** have the same variables with the same exponents: $3x^2$ and $-7x^2$ are like terms; $3x^2$ and $3x$ are not. Only like terms can be added or subtracted: $4a + 3b - a + 2b = 3a + 5b$.

Multiplying (expanding)

Type	Rule	Example
Monomial × binomial	Multiply each term inside	$3x(2x - 5) = 6x^2 - 15x$
Binomial × binomial	Each term × each term (FOIL)	$(x + 3)(x - 5) = x^2 - 5x + 3x - 15 = x^2 - 2x - 15$
Square of a binomial	$(a + b)^2 = a^2 + 2ab + b^2$	$(x + 4)^2 = x^2 + 8x + 16$
	$(a - b)^2 = a^2 - 2ab + b^2$	$(2x - 3)^2 = 4x^2 - 12x + 9$
Difference of squares	$(a + b)(a - b) = a^2 - b^2$	$(x + 7)(x - 7) = x^2 - 49$

Type	Rule	Example
Binomial × trinomial	Each term × each term	$(x + 2)(x^2 - x + 3) = x^3 + x^2 + x + 6$

Common mistake: $(x + 4)^2$ is **not** $x^2 + 16$ — don't forget the middle term.

Dividing

Divide each term by the monomial: $(12x^3 - 8x^2 + 4x) \div 4x = 3x^2 - 2x + 1$.

Factorising (the reverse of expanding)

Always look for a **common factor first**.

1. **Common factor:** $6x^2 + 9x = 3x(2x + 3)$; $5a^2b - 10ab^2 = 5ab(a - 2b)$.
2. **Difference of two squares:** $a^2 - b^2 = (a + b)(a - b)$.
   - $x^2 - 25 = (x + 5)(x - 5)$
   - $9y^2 - 16 = (3y + 4)(3y - 4)$
   - $2x^2 - 50 = 2(x^2 - 25) = 2(x + 5)(x - 5)$ (common factor first)
3. **Trinomial $x^2 + bx + c$:** find two numbers that **multiply to c** and **add to b**.
   - $x^2 + 7x + 12 \rightarrow 3 \times 4 = 12$, $3 + 4 = 7 \rightarrow (x + 3)(x + 4)$
   - $x^2 - x - 12 \rightarrow -4 \times 3 = -12$, $-4 + 3 = -1 \rightarrow (x - 4)(x + 3)$
   - $x^2 - 8x + 15 \rightarrow -3 \times -5 = 15$, $-3 + (-5) = -8 \rightarrow (x - 3)(x - 5)$

Check a factorisation by expanding it again.

Simplifying algebraic fractions

Factorise top and bottom, then cancel common **factors** (never cancel terms): $(x^2 - 9)/(x + 3) = (x + 3)(x - 3)/(x + 3) = x - 3$.

Solving equations

An equation is solved when you find the value of the variable that makes both sides equal. Whatever you do to one side, do to the other.

Type	Example	Solution
Linear	$3x - 7 = 11$	$3x = 18 \rightarrow$ **x = 6**
Variable on both sides	$5x + 2 = 2x - 10$	$3x = -12 \rightarrow$ **x = −4**
With brackets	$2(x - 3) = x + 4$	$2x - 6 = x + 4 \rightarrow$ **x = 10**
With fractions	$x/3 + x/2 = 5$	× LCD 6: $2x + 3x = 30 \rightarrow 5x = 30 \rightarrow$ **x = 6**
Product = 0	$(x - 2)(x + 5) = 0$	$x - 2 = 0$ or $x + 5 = 0 \rightarrow$ **x = 2 or x = −5**
Quadratic (factorise first)	$x^2 - 5x + 6 = 0$	$(x - 2)(x - 3) = 0 \rightarrow$ **x = 2 or 3**
Exponential	$2^x = 32$	$2^x = 2^5 \rightarrow$ **x = 5**

Check your answer by substituting it back into the original equation.

Word problems

1. Let x = the unknown.
2. Write an equation from the words.

3. Solve and answer in words.

Example: The sum of three consecutive numbers is 72. Let the numbers be x, x + 1, x + 2. Then 3x + 3 = 72 → x = 23. The numbers are **23, 24 and 25**.

Example: Thabo is 3 times as old as his sister. In 4 years the sum of their ages will be 32. Sister = x, Thabo = 3x → (x + 4) + (3x + 4) = 32 → 4x = 24 → x = 6. Sister is **6**, Thabo is **18**.

Patterns, functions and linear graphs

Number patterns

Linear pattern — a **constant difference** between terms. The general term is $\mathbf{T_n = dn + c}$, where d is the constant difference.

Example: 5; 8; 11; 14; ...

- Constant difference d = 3, so $T_n = 3n + c$.
- Use $T_1 = 5$: 3(1) + c = 5 → c = 2. So $\mathbf{T_n = 3n + 2}$.
- $T_{50} = 3(50) + 2 =$ **152**.
- Which term equals 89? 3n + 2 = 89 → 3n = 87 → **n = 29** (the 29th term).

Shortcut: c = T_1 − d (the "zero term").

Geometric pattern — a **constant ratio** (multiply by the same number): 3; 6; 12; 24; ... (× 2). $T_n = 3 \times 2^{n-1}$.

Patterns with pictures (e.g. matchsticks): count the items in each figure, find the constant difference, then the general rule. *Figure 1 = 4 sticks, figure 2 = 7, figure 3 = 10 → $T_n = 3n + 1$.*

Functions and relationships

A **function** gives exactly one output (y) for each input (x). It can be shown as a **flow diagram, table, formula (equation) or graph**.

x	−2	−1	0	1	2
y = 2x + 1	−3	−1	1	3	5

- **Input values** = x (independent variable). **Output values** = y (dependent variable).
- To find the rule from a table: the constant difference in y (when x goes up by 1) is the number in front of x; the y-value when x = 0 is the constant.

Linear graphs: y = mx + c

A linear function forms a **straight line**.

- **m = gradient (slope)** = steepness = rise ÷ run = $(y_2 - y_1) \div (x_2 - x_1)$.
  - m > 0: line goes **up** to the right. m < 0: line goes **down**. m = 0: horizontal line.
- **c = y-intercept** = where the line cuts the **y-axis** (put x = 0).
- **x-intercept** = where the line cuts the **x-axis** (put y = 0 and solve).

Example: y = 2x − 4

- Gradient = 2; y-intercept: x = 0 → y = −4 → point (0; −4).
- x-intercept: 0 = 2x − 4 → x = 2 → point (2; 0).
- **To draw it:** plot both intercepts and join them with a ruler, extending the line.

Example (gradient from two points): through (1; 3) and (4; 9): m = (9 − 3) ÷ (4 − 1) = 6 ÷ 3 = **2**.

Special lines

Equation	Line
y = k (e.g. y = 3)	**Horizontal** line through 3 on the y-axis; gradient 0
x = k (e.g. x = −2)	**Vertical** line through −2 on the x-axis; gradient undefined
y = mx (c = 0)	Passes through the **origin** (0; 0)

- **Parallel lines** have the **same gradient**: y = 3x + 1 and y = 3x − 5 are parallel.
- **Point of intersection** of two lines: set the equations equal and solve. y = x + 1 and y = −x + 5: x + 1 = −x + 5 → x = 2, y = 3 → **(2; 3)**.

Interpreting graphs

- **Increasing** (going up), **decreasing** (going down), **constant** (flat).
- **Linear** (straight) vs **non-linear** (curved).
- **Discrete** (separate points, e.g. number of people) vs **continuous** (joined line, e.g. time, temperature).
- On a distance–time graph, a steeper line means faster; a flat part means stopped.

Geometry, Pythagoras and transformations

In geometry questions, every statement needs a **reason** in brackets. Use the standard reasons below.

Angles and lines

Fact	Reason to write
Angles on a straight line add up to 180°	(∠s on a str line)
Angles around a point add up to 360°	(∠s round a pt)
Vertically opposite angles are equal	(vert opp ∠s)
Parallel lines: **corresponding** angles are equal (F-shape)	(corr ∠s; AB ∥ CD)
Parallel lines: **alternate** angles are equal (Z-shape)	(alt ∠s; AB ∥ CD)
Parallel lines: **co-interior** angles add up to 180° (U-shape)	(co-int ∠s; AB ∥ CD)

Types of angles: acute (< 90°), right (90°), obtuse (90°–180°), straight (180°), reflex (180°–360°), revolution (360°). **Complementary** angles add to 90°; **supplementary** angles add to 180°.

Triangles

- **Sum of angles** in a triangle = 180° (sum of ∠s of △).
- **Exterior angle** = sum of the two opposite interior angles (ext ∠ of △).
- **Isosceles** triangle: two equal sides, and the angles opposite them are equal (∠s opp = sides).
- **Equilateral** triangle: all sides equal, all angles 60°.
- **Scalene:** no equal sides. **Right-angled:** one 90° angle.

Congruent and similar triangles

Congruent (≡) = exactly the same shape and size. Conditions:

Condition	Meaning
SSS	Three sides equal
SAS	Two sides and the **included** angle equal
AAS (or SAA)	Two angles and a corresponding side equal
RHS	Right angle, hypotenuse and one other side equal

(AAA and SSA do **not** prove congruency.)

Similar (|||) = same shape, different size. Conditions: **all corresponding angles equal** (AAA), or **all corresponding sides in the same ratio**.

Example: △ABC ||| △DEF with AB = 4, DE = 8, BC = 5. Ratio = 8 ÷ 4 = 2, so EF = 5 × 2 = **10**.

Write the letters in **corresponding order**: △ABC ≡ △DEF means A matches D, B matches E, C matches F.

Quadrilaterals

Shape	Properties
Parallelogram	Opposite sides parallel and equal; opposite angles equal; diagonals bisect each other
Rectangle	Parallelogram with four right angles; diagonals equal
Rhombus	Parallelogram with four equal sides; diagonals bisect at 90° and bisect the angles
Square	All sides equal, all angles 90°; diagonals equal and bisect at 90°
Trapezium	One pair of opposite sides parallel
Kite	Two pairs of adjacent sides equal; diagonals perpendicular; one diagonal bisects the other

Sum of the angles in a quadrilateral = **360°**. Sum of interior angles of any polygon = **180°(n − 2)**, n = number of sides (hexagon: 180 × 4 = 720°).

Theorem of Pythagoras

In a **right-angled triangle**: **hypotenuse² = sum of the squares of the other two sides**: $c^2 = a^2 + b^2$. The hypotenuse (c) is the longest side, opposite the right angle.

Find the hypotenuse: sides 6 cm and 8 cm. $c^2 = 6^2 + 8^2 = 36 + 64 = 100 \rightarrow c =$ **10 cm**.

Find a shorter side: hypotenuse 13, one side 5. $a^2 = 13^2 - 5^2 = 169 - 25 = 144 \rightarrow a =$ **12**.

Is it right-angled? Sides 7, 24, 25: $7^2 + 24^2 = 49 + 576 = 625 = 25^2 \rightarrow$ **yes**.

Useful triples: 3-4-5, 5-12-13, 8-15-17, 7-24-25 (and multiples, e.g. 6-8-10).

Transformation geometry

Transformation	What happens	Rule on the Cartesian plane
Translation (slide)	Every point moves the same distance in the same direction	(x; y) → (x + a; y + b). 3 right, 2 down: (x; y) → (x + 3; y − 2)
Reflection in the x-axis	Mirror image across the x-axis	(x; y) → (x; −y)
Reflection in the y-axis	Mirror image across the y-axis	(x; y) → (−x; y)

Transformation	What happens	Rule on the Cartesian plane
Reflection in y = x	Swap x and y	(x; y) → (y; x)
Rotation 90° clockwise about the origin	Turn	(x; y) → (y; −x)
Rotation 90° anticlockwise about the origin	Turn	(x; y) → (−y; x)
Rotation 180° about the origin	Half turn	(x; y) → (−x; −y)
Enlargement / reduction by factor k from the origin	Shape gets bigger (k > 1) or smaller (0 < k < 1)	(x; y) → (kx; ky). Area is multiplied by $\mathbf{k^2}$

Translations, reflections and rotations give **congruent** shapes. Enlargements give **similar** shapes.

Example: A(2; 5) reflected in the y-axis → A′(−2; 5). Rotated 180° → A′(−2; −5). Enlarged by factor 3 → A′(6; 15).

Perimeter, area, surface area and volume

Perimeter is the distance around a shape (units: cm, m). **Area** is the space inside a flat shape (cm², m²). **Volume** is the space inside a 3D object (cm³, m³).

Area and perimeter of 2D shapes

Shape	Perimeter	Area
Square (side s)	4s	s^2
Rectangle (l × b)	2l + 2b	l × b
Triangle	sum of sides	½ × base × perpendicular height
Parallelogram	sum of sides	base × perpendicular height
Trapezium	sum of sides	½ × (sum of parallel sides) × height
Rhombus / kite	sum of sides	½ × diagonal_1 × diagonal_2
Circle (radius r)	**Circumference = 2πr** (or πd)	$\mathbf{\pi r^2}$

Use π ≈ 3.14 (or the calculator π key). Diameter = 2 × radius.

Example: circle with diameter 10 cm → r = 5. Circumference = 2 × 3.14 × 5 = **31.4 cm**; area = 3.14 × 5² = **78.5 cm²**.

Unit conversions

Length	Area	Volume / capacity
1 cm = 10 mm	1 cm² = 100 mm²	1 cm³ = 1 000 mm³
1 m = 100 cm	1 m² = 10 000 cm²	1 m³ = 1 000 000 cm³
1 km = 1 000 m		**1 cm³ = 1 mℓ**; 1 000 cm³ = 1 ℓ; 1 m³ = 1 000 ℓ (1 kℓ)

Surface area of 3D objects

Surface area = the total area of all the faces (the **net** of the object).

Object	Surface area
Cube (side s)	$6s^2$
Rectangular prism (l, b, h)	2(lb + lh + bh)
Triangular prism	2 × area of triangle + (perimeter of triangle × length)
Cylinder (radius r, height h)	$2\pi r^2 + 2\pi rh$ (two circles + the curved rectangle)

Example (rectangular prism 5 cm × 3 cm × 4 cm): SA = 2(5×3 + 5×4 + 3×4) = 2(15 + 20 + 12) = 2 × 47 = **94 cm²**.

Example (cylinder r = 7 cm, h = 10 cm, π = 3.14): SA = $2(3.14)(7^2) + 2(3.14)(7)(10)$ = 307.72 + 439.6 = **747.32 cm²**.

An **open** container (no lid) has one fewer face: open cylinder = $\pi r^2 + 2\pi rh$.

Volume of 3D objects

For any **prism**: **Volume = area of the base × height**.

Object	Volume
Cube	s^3
Rectangular prism	l × b × h
Triangular prism	(½ × base × height of triangle) × length of prism
Cylinder	$\pi r^2 \times h$

Example: rectangular box 5 cm × 3 cm × 4 cm: V = 5 × 3 × 4 = **60 cm³** = 60 mℓ.

Example: cylinder r = 7 cm, h = 10 cm: V = 3.14 × 49 × 10 = **1 538.6 cm³** ≈ 1.54 ℓ.

Example: triangular prism, triangle base 6 cm, triangle height 4 cm, prism length 10 cm: V = (½ × 6 × 4) × 10 = 12 × 10 = **120 cm³**.

Effect of changing dimensions

If all lengths are multiplied by k: **perimeter × k**, **area × k^2**, **volume × k^3**.

Example: doubling the sides of a cube makes its surface area 4 times bigger and its volume 8 times bigger.

Financial maths, data handling and probability

Financial maths

Term	Meaning
Principal (P)	Amount invested or borrowed at the start
Interest	Extra money earned (investment) or paid (loan)
Rate (i)	Interest rate per year, as a decimal (8% = 0.08)
n	Number of years
Accumulated amount (A)	Principal + interest at the end

Simple interest — interest is calculated on the **original amount only**: **$A = P(1 + i \times n)$**

Example: R5 000 at 8% p.a. simple interest for 3 years. $A = 5\,000(1 + 0.08 \times 3) = 5\,000 \times 1.24 =$ **R6 200**. Interest = R1 200.

Compound interest — interest is calculated on the **principal plus interest already earned**: **$A = P(1 + i)^n$**

Example: R5 000 at 8% p.a. compound interest for 3 years. $A = 5\,000(1.08)^3 =$ **R6 298.56**. Compound interest earns more than simple interest over time.

Hire purchase (HP) — buying on credit: pay a deposit, then monthly instalments with simple interest on the balance.

Example: A TV costs R12 000. Deposit 10%, balance paid over 2 years at 15% p.a. simple interest.

1. Deposit = 10% × 12 000 = R1 200 → balance = R10 800
2. Interest = 10 800 × 0.15 × 2 = R3 240 → amount owed = R14 040
3. Monthly instalment = 14 040 ÷ 24 = **R585**
4. Total paid = 1 200 + 14 040 = **R15 240** (R3 240 more than the cash price)

Other financial terms:

- **Profit** = selling price − cost price; **loss** if the selling price is lower. % profit = profit ÷ cost price × 100.
- **Discount** = reduction in price. 20% off R250 = R250 − R50 = R200.
- **VAT** in South Africa is **15%**. Price incl. VAT = price excl. VAT × 1.15. Price excl. VAT = price incl. VAT ÷ 1.15.
- **Inflation** = the general rise in prices over time (use the compound interest formula).
- **Exchange rates:** if \$1 = R18, then \$50 = 50 × 18 = R900, and R360 = 360 ÷ 18 = \$20.
- **Budget:** a plan of income and expenses. **Income** − **expenses** = savings (surplus) or a shortfall (deficit).

Data handling

The data cycle: pose a question → collect data (questionnaire, survey, observation, experiment) → organise (tally table, frequency table, stem-and-leaf) → represent (graphs) → analyse → interpret and report.

- **Population** = the whole group; **sample** = a part of the population that is studied. A sample must be **representative** and not **biased**.
- **Categorical** data (colours, sports) vs **numerical** data: **discrete** (counted, e.g. number of siblings) or **continuous** (measured, e.g. height).

Measures of central tendency and spread — data: 4; 7; 7; 9; 13

Measure	How to find it	Answer
Mean (average)	Sum ÷ number of values	40 ÷ 5 = **8**
Median	Middle value of the **ordered** data (even number of values: average of the two middle ones)	**7**
Mode	Value that occurs most often	**7**
Range	Highest − lowest	13 − 4 = **9**

An **outlier** is a value far from the rest; it affects the mean most, so the median is better when there are outliers.

Graphs:

- **Bar graph** — categorical data; gaps between bars.
- **Histogram** — grouped numerical data; no gaps.
- **Pie chart** — parts of a whole. Angle of a sector = (frequency ÷ total) × 360°.
- **Line graph** — changes over time.
- **Scatter plot** — relationship between two variables (positive, negative or no correlation).
- Watch for **misleading graphs**: axes not starting at zero, uneven scales, missing labels.

Probability

Probability = how likely an event is, from **0 (impossible)** to **1 (certain)**.

P(event) = number of favourable outcomes ÷ total number of possible outcomes

- A die: P(even number) = 3 ÷ 6 = ½; P(7) = 0.
- P(event does **not** happen) = 1 − P(event).
- A **sample space** lists all possible outcomes.

Two events together — use a table or tree diagram.

Example: Toss a coin and roll a die. Sample space = 2 × 6 = **12** outcomes (H1, H2, ..., T6). P(heads and a 6) = 1 ÷ 12.

Example: Toss two coins: HH, HT, TH, TT. P(one head and one tail) = 2 ÷ 4 = ½.

Theoretical vs relative frequency:

- **Theoretical probability** — what should happen (P(heads) = ½).
- **Relative frequency** — what actually happened in an experiment = number of times event occurred ÷ number of trials. 30 heads in 50 tosses → 30 ÷ 50 = 0.6.
- The more trials you do, the closer the relative frequency gets to the theoretical probability.